Cluster Naming Evaluation Report

OWNER: Toward Unsupervised Open-World Named Entity Recognition

Comparison of BERT-based and LLM-based (Llama-3.1-8B) Naming
Evaluated with BERTScore across Five CrossNER Domains

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Abstract

OWNER [1] organises extracted entities into clusters whose integer IDs carry no semantic meaning. This report evaluates two automatic cluster-naming strategies proposed in the paper: (1) **BERT-based naming**, which predicts the most frequent `[MASK]` token across all entities in a cluster, and (2) **LLM-based naming** (Llama-3.1-8B via Ollama), which infers a type label from a sample of n=16 entities using the Figure 3 prompt. Named clusters are scored against ground-truth entity type labels using **BERTScore** (RoBERTa-large backbone) across five CrossNER domains: AI, Literature, Music, Politics, and Science. BERT naming achieves an average F_1 of **0.966** while LLM naming achieves **0.944**, with BERT performing consistently better except on the Music domain where both methods are within 0.001 F_1 of each other.

1 Introduction

Named Entity Recognition (NER) traditionally requires a fixed, predefined set of entity types. OWNER removes this constraint by clustering entity embeddings into semantically coherent groups, automatically discovering the number and nature of entity types without annotation. However, the resulting clusters are identified only by integer IDs. Assigning meaningful names to these clusters is necessary for interpretability, knowledge graph construction, and human evaluation.

The paper proposes two complementary naming strategies (Section III-B-3-C):

- **BERT naming**: leverage the same BERT encoder used for entity typing. For each entity, the prompt $\mathcal{P}(e, X)$ = `‘‘{sentence} {entity} is a [MASK].’’` is passed through the MLM head; the top-1 predicted token is collected. The most frequent predicted token across all entities in a cluster becomes its name. This approach is cost-free as the encoder is already available.
- **LLM naming**: a random sample of $n = 16$ entity strings is presented to Llama-3.1-8B-Instruct (via Ollama) using the prompt in Figure 3 of the paper (temperature = 0). The model responds with a single entity type name.

Both approaches are evaluated against ground-truth labels using BERTScore [2], which measures token-level semantic similarity via contextualised RoBERTa-large embeddings.

2 Experimental Setup

2.1 Datasets

We evaluate on five CrossNER target domains used in the OWNER paper. Table 1 summarises the number of entity types and aligned clusters in each domain.

Table 1: CrossNER target domains and cluster statistics.

Domain	**True entity types k**	**Aligned clusters**
AI	14	16
Literature	12	14
Music	13	20
Politics	9	22
Science	17	20

2.2 Ground-Truth Alignment

The ground-truth label for each cluster is determined by finding the true entity type that contributes the highest proportion of entities to that cluster (column-wise argmax of the confusion matrix), stored as `et_test_cluster_naming_gt.json`. Only clusters present in all three files (GT, BERT, LLM) are included in the evaluation.

2.3 BERTScore

BERTScore computes precision, recall, and F_1 by greedily matching tokens of the candidate string to tokens of the reference string using cosine similarity of RoBERTa-large contextual embeddings. We report macro-averaged P, R, F_1 across all aligned clusters within each domain, and then macro-average over domains.

3 Results

3.1 BERTScore per Domain

Table 2 presents the full BERTScore results. Bold entries indicate the better F_1 for each domain.

Table 2: BERTScore (Precision / Recall / F_1) of BERT-based and LLM-based naming against ground-truth labels. k = number of aligned clusters. Best F_1 per domain in **bold**.

	BERT Naming				**LLM Naming (Llama-3.1-8B)**			
Domain	k	P	R	F_1	k	P	R	F_1
AI	16	0.992	0.986	**0.989**	16	0.915	0.945	0.929
Literature	14	0.974	0.975	**0.974**	14	0.950	0.960	0.954
Music	20	0.965	0.951	**0.958**	20	0.960	0.955	0.957
Politics	22	0.968	0.962	**0.964**	22	0.945	0.961	0.953
Science	20	0.949	0.943	**0.945**	20	0.922	0.933	0.926
Average	92	**0.970**	**0.963**	**0.966**	92	0.938	0.951	0.944

3.2 Qualitative Examples

Table 3 shows representative cluster naming examples from the five domains, illustrating where each method succeeds or struggles.

Table 3: Selected cluster naming examples. Green = correct or near-correct; red = incorrect.

Domain	Cluster	Ground Truth	BERT Name	LLM Name
AI	11	conference	conference	Conference
AI	9	university	university	Universities
AI	1	researcher	physicist	Computer Scientist
AI	13	person	vegetarian	Person
Literature	2	writer	poet	Author
Literature	13	literarygenre	genre	Literary work
Literature	0	book	vegetarian	Novel
Music	3	band	band	Band
Music	7	musicgenre	genre	Music genres
Music	4	musicalartist	vegetarian	Singer
Politics	7	politicalparty	party	Political party
Politics	11	politician	democrat	Politician
Politics	16	person	vegetarian	Actor
Science	15	protein	protein	Protein
Science	9	scientist	physicist	Physicist
Science	5	astronomicalobj	planet	Planet
Science	14	person	vegetarian	Actor

4 Analysis

4.1 BERT Naming Strengths and Limitations

BERT naming achieves the highest average F_1 (0.966) at zero additional inference cost since the encoder is already loaded for entity typing. It excels at domain-specific technical terms (*conference*, *university*, *protein*, *genre*) where BERT's pretraining vocabulary is well-aligned with the entity types.

However, BERT naming exhibits a known artefact: `vegetarian` appears repeatedly across domains as a predicted token. This occurs because the prompt template `‘‘X is a [MASK].’’` in unusual entity contexts triggers BERT to predict words from common fill-in-the-blank patterns unrelated to entity types. This affects clusters 13 (AI), 0 (Literature), 1 and 4 and 19 (Music), 13 and 16 (Politics), and 14 (Science).

An additional limitation is that BERT predicts a *single sub-word token*, so multi-word entity types such as `astronomicalobject`, `chemicalelement`, or `musicalartist` can never be predicted exactly.

4.2 LLM Naming Strengths and Limitations

LLM naming (Llama-3.1-8B, $n = 16$) produces more natural and multi-word type names (*Computer Scientist*, *Political party*, *Scientific Journals*), avoiding the single-token constraint. It never produces nonsensical artefacts like `vegetarian`.

However, the LLM occasionally over-generalises:

- *Stadiums* for `country` (AI, cluster 4)
- *Actor*/*Celebrity* for `person` (Politics, Science)
- *Song* for `album` (Music, cluster 6)

This occurs when the sampled entities are dominated by false positives or semantically diverse spans that mislead the model.

4.3 Domain-level Observations

- **AI** shows the largest gap between methods ($\Delta F_1 = 0.060$), likely because BERT's pretraining includes substantial AI literature where *conference* and *university* appear in exactly the fill-in-the-blank pattern.
- **Music** shows the smallest gap ($\Delta F_1 = 0.001$), with both methods performing well on concrete musical concepts.
- **Science** yields the lowest scores for both methods, reflecting the domain's fine-grained types (*chemicalelement*, *astronomicalobject*, *academicjournal*) that require specialised vocabulary.

5 Conclusion

Both naming strategies produce semantically meaningful cluster labels without any labelled data. BERT naming is the stronger method on average (F_1 0.966 vs 0.944) and comes at no extra cost, while LLM naming provides more readable, multi-word labels and avoids degenerate predictions. The two approaches are complementary: BERT naming is recommended for technical domains with single-word type names, while LLM naming is preferred when human-readable output or multi-word labels are required.

Future work could combine both approaches: use BERT naming as a candidate and LLM naming as a verifier, or fine-tune the LLM on domain-specific entity samples to reduce hallucination.

References

[1] P.-Y. Genest, P.-E. Portier, E. Egyed-Zsigmond, and M. Lovisetto, "OWNER – Toward Unsupervised Open-World Named Entity Recognition," *IEEE Access*, vol. 13, pp. 50077–50096, 2025.

[2] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating Text Generation with BERT," in *Proc. ICLR*, 2020.